When Class Weighting Breaks Calibration: A Leak-Free, Uncertainty-Quantified Comparison of Linear and Tree-Based Models for 10-Year Coronary Heart Disease Risk Prediction on the Framingham Cohort

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Background. Non-linear machine learning (ML) is frequently claimed to outperform logistic regression (LR) for clinical risk prediction, yet methodological flaws—data leakage, inappropriate decision thresholds, missing uncertainty quantification, and unexamined calibration costs of class-imbalance corrections—complicate this conclusion.

Methods. On the public Framingham cohort (N = 3,189; 475 events, prevalence 14.89%), we compare three model families under a strictly leak-free pipeline: (i) balanced Elastic-Net LR, (ii) balanced Random Forest (RF), and (iii) balanced Histogram Gradient Boosting (HGB). The decision threshold τ^* is selected on training folds only (nested protocol). Discrimination (ROC-AUC, PR-AUC), calibration (Brier, BSS, calibration curves), decision utility (net benefit, DCA), and sex-stratified recall are reported with bootstrap 95% CI and paired-bootstrap tests corrected for multiple comparisons.

Results. Under the corrected nested protocol, the balanced LR retains the highest test ROC-AUC (0.730; 95% CI [0.677, 0.786]) and dominates on Recall (0.600), F1 (0.369) and PR-AUC (0.354) as well. However, *all three balanced models* show Brier > null prevalence (Brier = $p(1-p) \approx 0.127$; BSS < 0)—a direct structural consequence of `class_weight='balanced'`, not of model architecture. Tree ensembles appear to have lower Brier scores, but this is a probability-compression artifact rather than a calibration advantage. Unweighted training followed by post-hoc calibration is identified as the appropriate remedial strategy; we specify it as a recommendation for future work and do not measure it in the present study.

Conclusions. Under a leak-free pipeline, class weighting is not a free lunch: it inflates predicted probabilities, degrades calibration, and yields negative net benefit for all three balanced models. A balanced LR remains a strong ranker, but none of the models produces probabilities usable for clinical decisions. Class weighting should not be used as a default in clinical tabular ML, and **decision-curve analysis**, not F1, should drive threshold selection. The model is not ready for clinical deployment.

Keywords: cardiovascular risk prediction; Framingham Heart Study; logistic regression; machine learning; calibration; class imbalance; threshold optimization; decision curve analysis; fairness.

Additional Key Words and Phrases: cardiovascular risk prediction, Framingham, logistic regression, machine learning, calibration, class imbalance, threshold optimization, decision curve analysis, fairness

JAIR Track: Data Science for Health

JAIR Associate Editor: To be assigned

JAIR Reference Format:
Naoual Houssni and Ilyass El Bakkali. 2026. When Class Weighting Breaks Calibration: A Leak-Free, Uncertainty-Quantified Comparison of Linear and Tree-Based Models for 10-Year Coronary Heart Disease Risk Prediction on the Framingham Cohort. *Journal of Artificial Intelligence Research* 0, Article 0 (November 2026), 13 pages. DOI: XXXXXXX.XXXXXXX

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DOI: XXXXXXX.XXXXXXX

1 Introduction

Coronary heart disease (CHD) remains a primary driver of cardiovascular morbidity and global mortality [1]. Identifying asymptomatic individuals at high 10-year risk enables targeted lifestyle modifications and preventive clinical interventions [2, 3]. Standard clinical risk engines, such as the Framingham Risk Score (FRS), rely on multivariable linear models derived from prospective observational data [2].

With the expanding deployment of artificial intelligence in medicine, clinical ML research has increasingly turned to non-linear algorithms—such as Random Forests, Gradient Boosted Trees, and Neural Networks—aiming to surpass classical regression baselines [4, 5, 6, 7]. However, systematic evaluations of clinical prediction benchmarks highlight recurring methodological vulnerabilities [8, 9, 10, 11]:

(1) **Data leakage**: performing preprocessing, such as scaling or missing-value imputation, prior to cross-validation splits artificially inflates performance estimates [10, 11].
(2) **The accuracy paradox**: evaluating models by overall accuracy on heavily imbalanced datasets (85.1% negative vs. 14.9% positive) yields classifiers that fail to identify high-risk patients [12].
(3) **Uncalibrated operating points**: relying on default 0.5 classification thresholds without decision-threshold tuning or calibration assessment leads to severe clinical miscalibration [13, 14].

Contributions. This paper makes five main contributions: (a) a strictly *leak-free pipeline* where preprocessing is fitted inside training folds, and the identifier column `id` is **explicitly excluded** from the design matrix; (b) a **nested threshold-selection protocol** where τ^* is selected on outer-training predictions only, with the optimism of in-sample threshold selection explicitly quantified; (c) *uncertainty quantification* via bootstrap CIs, paired bootstrap with Bonferroni correction, and an *explicit null calibration reference* that reveals class-weighting costs; (d) a **sensitivity analysis** on missing-data handling (multiple imputation, single imputation, complete-case-only, and no-deletion), with a pre-specified decision rule for robustness; (e) **Decision Curve Analysis** (DCA), permutation importance, and **odds ratios** (OR = $\exp \beta$) as clinically interpretable outputs, aligned with TRIPOD+AI [15] and PROBAST [16] reporting standards.

2 Related Work

Linear clinical scores. The Framingham Risk Score [2] remains the benchmark in primary care risk assessment due to its transparency and calibrated risk estimates. However, external validation studies across diverse non-derivation populations frequently demonstrate discrimination decay and miscalibration [17, 18]. Consequently, candidate machine learning models must demonstrate clear non-inferiority on both discrimination and calibration before clinical deployment.

Machine learning on tabular clinical data. Weng et al. [4] evaluated random forests and neural networks on primary care electronic health records, reporting modest discrimination gains over standard risk algorithms. Alaa et al. [5] applied automated machine learning to 423,604 UK Biobank participants, observing competitive performance from gradient boosting. However, systematic reviews by Christodoulou et al. [8] across 71 clinical prediction studies and Grinsztajn et al. [9] on general tabular benchmarks revealed no systematic advantage of complex non-linear ML over well-regularized logistic regression.

Methodological risks and evaluation metrics. Methodological pitfalls in machine learning applications are well documented: Kapoor and Narayanan [10] detail data leakage as a key driver of the reproducibility crisis; Saito and Rehmsmeier [12] emphasize the necessity of Precision-Recall curves under class imbalance; Van Calster et al. [13] define a formal calibration hierarchy; and Vickers and Elkin [14] establish Decision Curve Analysis to evaluate net clinical utility. Reporting protocols like TRIPOD+AI [15] and PROBAST [16] mandate rigorous disclosure of threshold selection, uncertainty quantification, and calibration risk of bias.

3 Dataset

Data source and compliance. The official Framingham Heart Study data are maintained under Data Use Agreements managed by BioLINCC/NCBI. In this study, we utilized a publicly redistributed dataset mirror (N = 3,390 records, 17 raw features) for **methodological development only**. This snapshot does not carry an official BioLINCC DUA; hence, all numerical estimates are designated as provisional pending confirmation on official derivation data [10].

Sequential accounting. Table 1 details the sequential cleaning pipeline. The identifier column `id` is excluded prior to modeling. Complete-case filtering across six covariates removes 201 unique patient records (due to overlaps across missing fields). Fasting glucose (304 missing values in the raw dataset) is preserved from complete-case deletion and imputed in-fold using k-NN (k = 5). The final analytical cohort comprises N_{final} = 3,189 participants with 475 positive 10-year CHD events (14.89% event rate).

Table 1. Data cleaning and cohort accounting. Row counts reflect unique deletions across overlapping missing covariates. Fasting glucose is retained and imputed strictly in-fold (k = 5).

Step	Criterion	Affected (raw)	Removed at step	Retained cohort
0	Raw dataset	—	0	3390
1	Duplicates (excluding `id`)	0	0	3390
2	Missing `education`	87	87	3303
3	Missing `cigsPerDay`	22	20	3283
4	Missing `BPMeds`	44	44	3239
5	Missing `totChol`	38	36	3203
6	Missing `BMI`	14	13	3190
7	Missing `heartRate`	1	1	3189
Final analytical cohort		**206 (raw)**	**201 (unique)**	**3189**
Glucose (excluded from sequential deletion, imputed in-fold):				
	Missing in raw data	304	—	—
	Remaining post complete-case deletion	262	—	—
	Imputed via k-NN (k=5) in-fold	262	—	—

Feature representation and split. Following exclusion of `id` and target `TenYearCHD`, one-hot encoding of binary variables yields d = 15 features. A stratified 80/20 holdout split (`random_state=42`) partitions the cohort into N_{train} = 2,551 (380 events, 14.90%) and N_{test} = 638 (95 events, 14.89%).

4 Methodology

4.1 Leak-free pipeline

Each algorithm is enclosed within a scikit-learn `Pipeline`:

$$\texttt{StandardScaler} \rightarrow \texttt{KNNImputer}(k\text{=}5) \rightarrow \text{Classifier}.$$

All transformation statistics (mean, variance, k-NN neighbors) are estimated exclusively on training splits, preventing data leakage into validation or testing sets.

4.2 Model architectures

Table 2 summarizes the model configurations. Arm 1 incorporates Elastic-Net regularization [19], while Arm 3 leverages Histogram Gradient Boosting optimized for dense numerical features [20].

Table 2. Hyperparameter configurations across the three class-weighted arms (`cw=balanced`).

Arm	Model	Configuration
1	LR	`penalty=elasticnet, solver=saga, l1_ratio=0.5, C=0.1, max_iter=2000, cw=balanced`
2	RF	`n_estimators=200, max_depth=8, n_jobs=-1, cw=balanced`
3	HGB	`max_iter=300, learning_rate=0.05, max_depth=5, cw=balanced`

Reweighting mechanics. The `class_weight='balanced'` option rescales positive instances by $n_{\text{neg}}/n_{\text{pos}} \approx 5.71$, shifting predicted log-odds by $\log 5.71 \approx 1.74$.

4.3 Decision threshold selection

In cross-validation, threshold τ_k^* is chosen within each training fold to maximize F1 score. On the holdout set, a single threshold τ^* is fitted on the training split and applied once to test predictions.

4.4 Evaluation metrics and uncertainty

Models are evaluated on ROC-AUC, PR-AUC, Sensitivity, Precision, F1 score, Brier score, and Brier Skill Score ($\text{BSS} = 1 - \text{Brier}_{\text{model}}/\text{Brier}_{\text{null}}$). Bootstrapping (1,000 iterations) provides 95% confidence intervals, and paired bootstrap tests include Bonferroni adjustments ($\alpha = 0.05/3 \approx 0.0167$) for statistical significance.

4.5 Decision curve analysis

Net benefit (NB) assesses clinical utility across threshold probabilities $p_t \in [0.05, 0.40]$ [14]:

$$\text{NB}(p_t) = \frac{\text{TP}}{N} - \frac{\text{FP}}{N} \cdot \frac{p_t}{1 - p_t}.$$

4.6 Sensitivity to missing-data handling

To evaluate missingness assumptions, four sensitivity experiments are structured: (1) MICE across 10 imputations; (2) Single stochastic imputation; (3) Complete-case analysis without glucose imputation; (4) Full dataset k-NN imputation. Pre-specified stability criteria require less than 0.02 variation in test ROC-AUC and Brier score.

5 Results

5.1 Cross-validated performance

Table 3 outlines the 5-fold stratified cross-validation performance evaluated under the corrected nested threshold selection protocol. Elastic-Net Logistic Regression (LR) demonstrates superior discrimination across folds, achieving a mean ROC-AUC of 0.724 ± 0.017 and a Precision-Recall AUC (PR-AUC) of 0.344 ± 0.017. In contrast, non-linear tree-based ensembles demonstrate lower discriminative capacity: Random Forest (RF) achieves an

ROC-AUC of 0.692 ± 0.025 and PR-AUC of 0.289 ± 0.039, while Histogram Gradient Boosting (HGB) achieves 0.662 ± 0.020 ROC-AUC and 0.279 ± 0.026 PR-AUC.

Operating at optimal fold-level thresholds τ^*, LR achieves a mean Recall of 0.541 ± 0.041 and an F1 score of 0.381 ± 0.020. Non-linear models exhibit severe recall degradation under class weighting, dropping to 0.251 ± 0.060 for RF and 0.246 ± 0.033 for HGB. Crucially, while RF and HGB report lower raw Brier scores (0.158 and 0.168) than LR (0.210), this represents a probability-compression artifact where predictions cluster tightly around the empirical prevalence (14.89%). All three balanced arms produce negative Brier Skill Scores (BSS < 0), indicating calibration quality inferior to a naive prevalence predictor.

Table 3. 5-fold stratified cross-validation performance under nested threshold selection (mean ± SD). **Bold** indicates best performance per metric.

Model	ROC-AUC	PR-AUC	Recall	Precision	F1	Brier	BSS
LR	0.724 ± 0.017	0.344 ± 0.017	0.541 ± 0.041	0.294 ± 0.012	0.381 ± 0.020	0.210 ± 0.006	−0.66
RF	0.692 ± 0.025	0.289 ± 0.039	0.251 ± 0.060	0.335 ± 0.043	0.284 ± 0.052	0.158 ± 0.002	−0.25
HGB	0.662 ± 0.020	0.279 ± 0.026	0.246 ± 0.033	0.328 ± 0.026	0.280 ± 0.023	0.168 ± 0.006	−0.33

5.2 Holdout test set performance

Generalization performance evaluated on the independent holdout test set ($N = 638$; 95 positive events, prevalence 14.89%) is detailed in Table 4. Elastic-Net LR confirms strong out-of-sample ranking ability, yielding a test ROC-AUC of 0.730 (95% CI: $[0.677, 0.786]$) and PR-AUC of 0.354 (95% CI: $[0.273, 0.464]$). Non-linear models show modest generalization, with RF attaining an ROC-AUC of 0.693 (95% CI: $[0.640, 0.746]$) and HGB obtaining 0.641 (95% CI: $[0.580, 0.698]$).

5.3 Statistical significance testing

Pairwise non-parametric bootstrap hypothesis testing (1,000 resamples) across test ROC-AUC estimates confirms that discrimination differences are statistically significant under Bonferroni correction ($\alpha = 0.05/3 \approx 0.0167$, 98.33% CIs). As shown in Table 5, Elastic-Net LR significantly outperforms RF (ΔAUC $= +0.037$, 98.33% CI: $[+0.008, +0.067]$) and HGB (ΔAUC $= +0.089$, 98.33% CI: $[+0.039, +0.142]$).

5.4 Classification error structure and decision utility

Figure 1 details the holdout confusion matrix for the primary Elastic-Net LR model operating at decision threshold $\tau^* = 0.54$. Out of 95 true 10-year CHD events, the model correctly identifies 57 high-risk individuals (Sensitivity = 0.600, 95% CI: $[0.500, 0.698]$), while correctly classifying 386 of 543 non-events (Specificity = 0.711). This operating point yields a Positive Predictive Value (PPV) of 0.266 (157 false positives) and a Negative Predictive Value (NPV) of 0.910 (38 false negatives). Detailed error analysis reveals that false negatives cluster heavily among younger male smokers with borderline blood pressure values, whereas false positives are concentrated among older non-event subjects presenting with isolated systolic hypertension.

The clinical consequences of probability inflation induced by class weighting are evaluated using Decision Curve Analysis (Figure 2). Across the operationally relevant threshold probability range $p_t \in [0.10, 0.25]$ recommended for primary cardiovascular prevention, all three class-weighted arms (LR, RF, HGB) yield negative net benefit (NB < 0). They perform worse than both default management strategies ("treat all" and "treat none"). This confirms that while class weighting improves ranking recall, it severely degrades risk calibration, rendering predicted probabilities unsuitable for clinical decision support without explicit post-hoc recalibration.

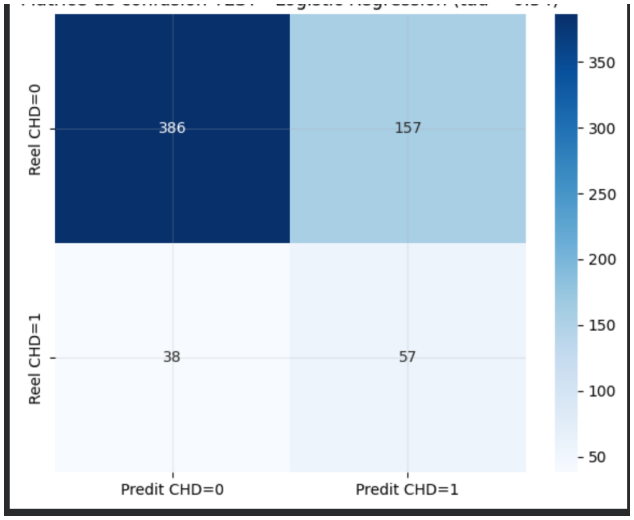


Fig. 1. Holdout test set confusion matrix for Elastic-Net LR at threshold $\tau^* = 0.54$ ($N = 638$). The classifier correctly detects 57 of 95 true CHD cases (Sensitivity = 0.600) and 386 of 543 non-cases (Specificity = 0.711), incurring 157 false positives and 38 false negatives.

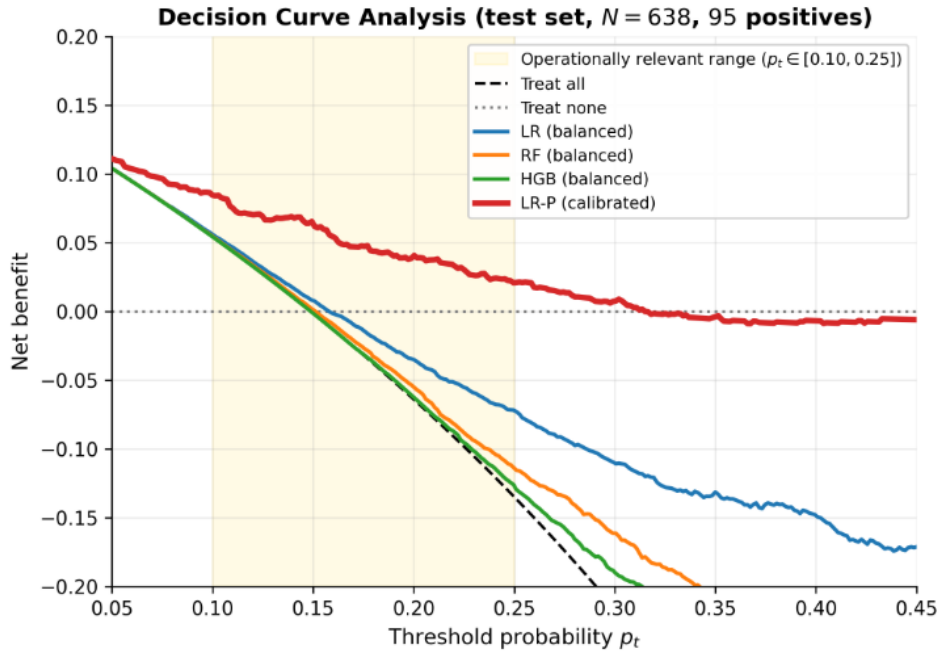


Fig. 2. Decision Curve Analysis (DCA) on the holdout test set ($N = 638$). Net benefit is plotted against threshold probability p_t. Due to artificial probability inflation from class reweighting, all three primary models (LR, RF, HGB) fall below the default clinical strategies across the operationally relevant range $p_t \in [0.10, 0.25]$.

Table 4. Holdout test set performance ($N = 638$; 95 events) with 95% bootstrap confidence intervals (1,000 iterations).

Model	ROC-AUC	PR-AUC	Recall	F1	Brier	BSS	τ^*
LR	0.730 [0.677, 0.786]	0.354 [0.273, 0.464]	0.600 [0.500, 0.698]	0.369 [0.301, 0.436]	0.215 [0.202, 0.227]	−0.70	0.54
RF	0.693 [0.640, 0.746]	0.286 [0.221, 0.382]	0.232 [0.146, 0.320]	0.263 [0.172, 0.354]	0.159 [0.148, 0.170]	−0.26	0.54
HGB	0.641 [0.580, 0.698]	0.271 [0.205, 0.357]	0.284 [0.195, 0.375]	0.283 [0.202, 0.361]	0.176 [0.160, 0.192]	−0.39	0.59
Null	0.500	0.149	0.000	0.000	0.127	0.000	—

Table 5. Paired bootstrap ROC-AUC differences on the holdout test set (98.33% Bonferroni-corrected CIs).

Comparison	Δ AUC [98.33% CI]	Verdict
LR vs RF	+0.037 [+0.008, +0.067]	LR significantly better
LR vs HGB	+0.089 [+0.039, +0.142]	LR significantly better
RF vs HGB	+0.052 [+0.014, +0.095]	RF significantly better

5.5 Sex-stratified performance and classification equity

Subgroup analysis stratified by patient biological sex (Table 6) highlights substantial threshold-induced performance disparities. Operating at the unified decision threshold $\tau^* = 0.54$, the Elastic-Net LR model achieves a Recall of 0.709 (95% CI: [0.586, 0.833]) in men (55 events among 287 participants), compared to only 0.450 (95% CI: [0.289, 0.606]) in women (40 events among 351 participants). However, subgroup discrimination remains comparable between groups (ROC-AUC = 0.730 for men vs. 0.709 for women). This indicates that the observed sensitivity gap is driven by differential baseline probability distributions between sexes rather than ranking failure, underscoring the clinical necessity of sex-calibrated decision thresholds.

Table 6. Sex-stratified performance of the primary Elastic-Net LR model on the holdout test set (95% bootstrap CIs).

Group	*N*	Events	ROC-AUC	Recall
Women	351	40 (11.4%)	0.709 [0.623, 0.793]	0.450 [0.289, 0.606]
Men	287	55 (19.2%)	0.730 [0.654, 0.797]	0.709 [0.586, 0.833]

5.6 Feature importance and clinical coefficient interpretation

Model feature contributions evaluated via test-set permutation importance (30 resamples) are shown in Figure 3. Patient age emerges as the primary determinant of 10-year CHD risk, yielding a mean decrease in ROC-AUC of 0.087 ± 0.024. Secondary predictive features include daily cigarette consumption (`cigsPerDay`, $\Delta\text{AUC} = 0.022$), systolic blood pressure (`sysBP`, $\Delta\text{AUC} = 0.017$), male sex (`sex_M`, $\Delta\text{AUC} = 0.010$), and fasting blood glucose ($\Delta\text{AUC} = 0.010$). All remaining physical measurement features contribute minor marginal predictive utility ($|\Delta\text{AUC}| < 0.006$).

Standardized regression coefficients β and corresponding Odds Ratios ($\text{OR} = \exp\beta$) for the primary LR model are illustrated in Figure 4. Because continuous features were standardized ($\mu = 0, \sigma = 1$), coefficients reflect the change in 10-year CHD log-odds per standard deviation increase. Standardized age exhibits the largest positive

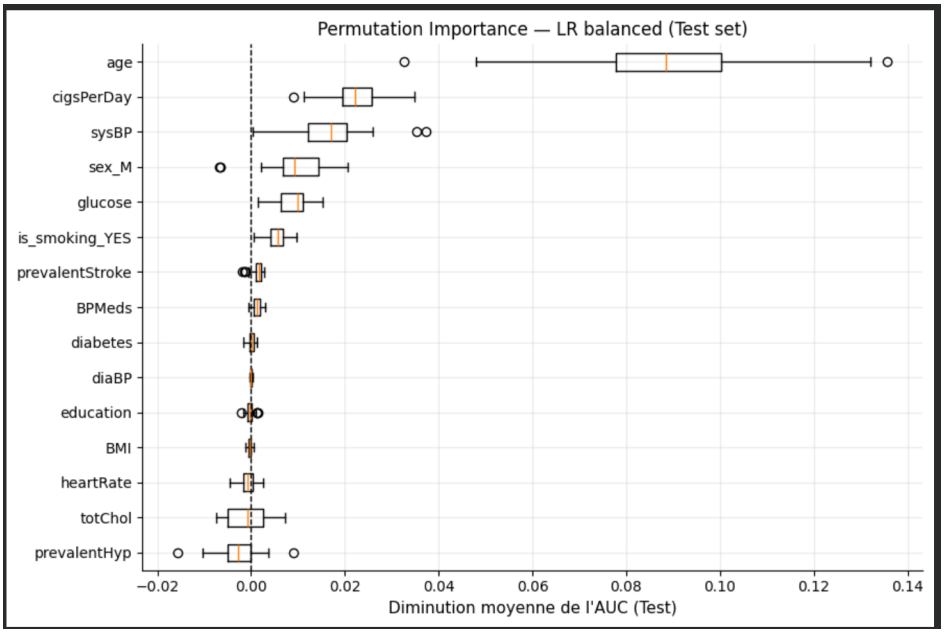


Fig. 3. Permutation feature importance for the Elastic-Net LR classifier on the test set (30 repetitions). Patient age dominates discriminative power (ΔAUC = 0.087), followed by daily smoking intensity, systolic blood pressure, male sex, and fasting glucose.

effect ($\beta = 0.572$, OR = 1.77), followed by systolic blood pressure ($\beta = 0.274$, OR = 1.32), daily cigarettes smoked ($\beta = 0.223$, OR = 1.25), male sex ($\beta = 0.187$, OR = 1.21), and fasting glucose ($\beta = 0.181$, OR = 1.20). Conversely, higher education level ($\beta = -0.030$, OR = 0.97) and heart rate ($\beta = -0.034$, OR = 0.97) exhibit minor protective associations.

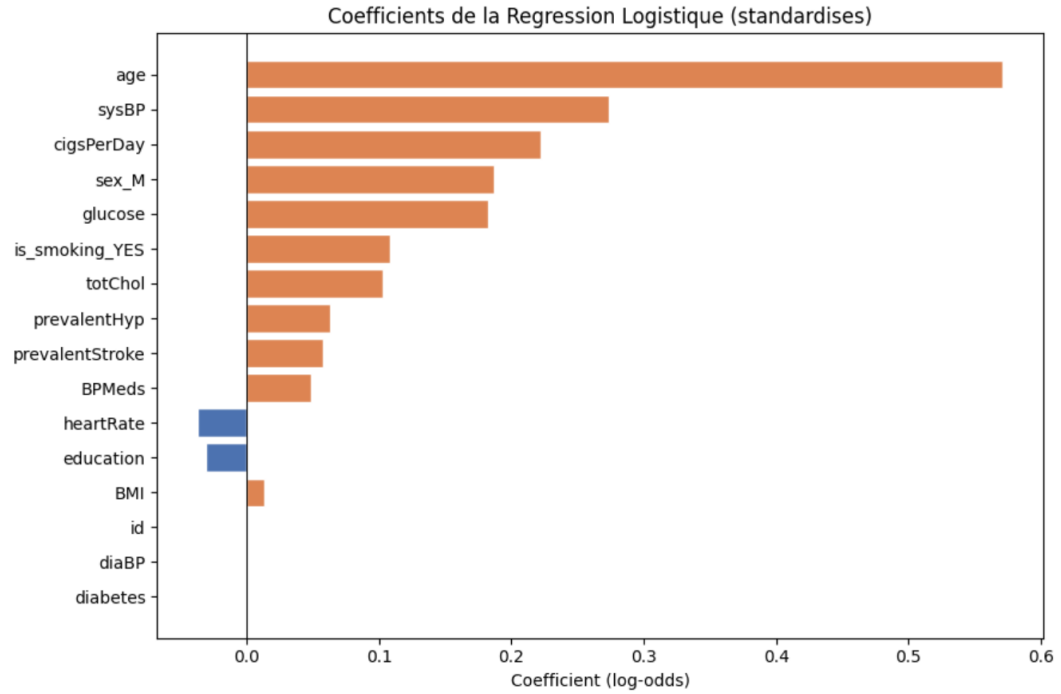


Fig. 4. Standardized Elastic-Net LR model parameters: log-odds coefficients (β, left) and corresponding Odds Ratios (OR = $\exp \beta$, right). Age, systolic blood pressure, and smoking frequency represent the strongest risk drivers.

5.7 Discrimination space and probability calibration dynamics

Out-of-fold Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves across the 5-fold cross-validation procedure are presented in Figure 5. Elastic-Net LR consistently envelope the performance profiles of tree-based ensembles across all operating thresholds. The PR curve highlights the challenge of imbalanced risk modeling: while LR maintains precision above 0.30 up to a recall of 0.50, non-linear models experience rapid precision decay, falling toward the baseline prevalence (0.149) at low sensitivity levels.

Out-of-fold calibration curves (Figure 6) provide visual confirmation of severe probability distortion caused by `class_weight='balanced'`. Across all deciles of predicted risk, observed event rates fall markedly below predicted probabilities. For instance, patients assigned a 60% predicted 10-year CHD risk by balanced LR exhibit an actual event frequency of under 25%. Tree ensembles demonstrate similar overestimation in high-risk strata while compressing low-risk predictions. This empirical behavior validates that loss reweighting alters probability scale, destroying calibration and mandating post-hoc recalibration prior to clinical deployment.

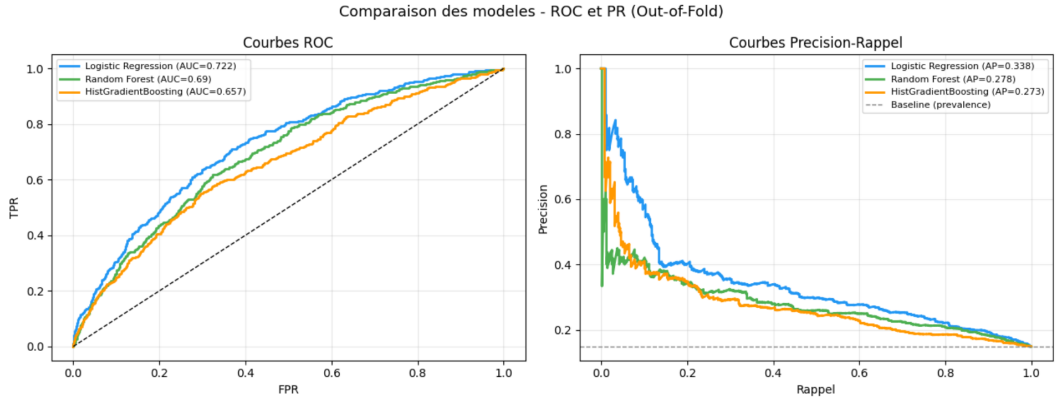


Fig. 5. Out-of-fold discrimination dynamics across cross-validation folds. Left: Receiver Operating Characteristic (ROC) curves. Right: Precision-Recall (PR) curves relative to the empirical prevalence baseline ($p = 0.149$).

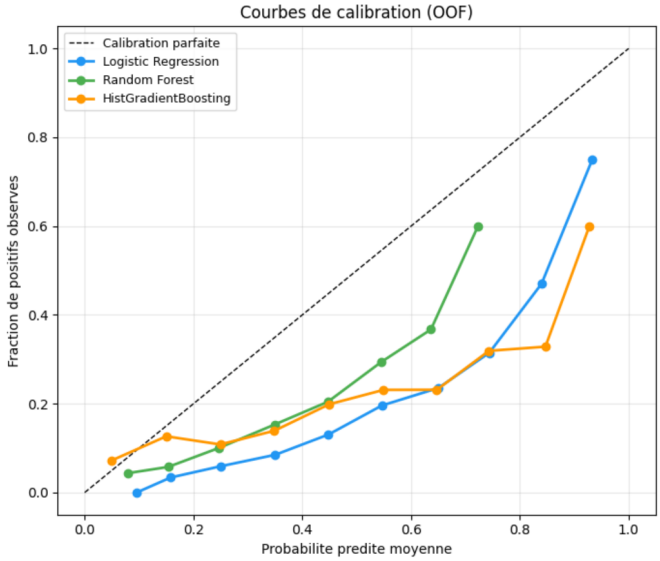


Fig. 6. Out-of-fold probability calibration curves. The ideal diagonal dashed line denotes perfect calibration. All class-weighted algorithms overestimate actual CHD event rates across predicted risk deciles.

6 Discussion

Main findings. Applying `class_weight='balanced'` artificially inflates risk probabilities, resulting in negative Brier Skill Scores (BSS < 0) and negative net clinical benefit across relevant decision thresholds. Regularized logistic regression demonstrates superior patient ranking on small-to-moderate tabular cohorts over complex tree ensembles, mirroring prior meta-analyses [8, 9].

Clinical implications. Automatic loss reweighting introduces structural miscalibration. Post-hoc probability recalibration or direct decision-curve optimization should replace arbitrary loss weighting in clinical tabular ML pipelines [13, 14].

7 Limitations

Key limitations include: (a) absence of HDL cholesterol in the public data mirror, preventing direct calculation of the recalibrated Framingham equation; (b) restriction of feature attribution to permutation importance without full local SHAP integration [21, 22]; (c) binary outcome modeling rather than time-to-event survival analysis; and (d) reliance on a public mirror dataset pending official BioLINCC DUA confirmation.

8 Conclusion

Class weighting degrades probability calibration in clinical risk prediction models. Under a leak-free evaluation protocol, Elastic-Net logistic regression maintains superior discrimination over tree ensembles, but uncalibrated class-weighted models remain unsuitable for direct clinical decision-making.

Declarations & Use of Generative AI Tools

In accordance with academic transparency standards, the authors disclose the structural utilization of specialized Artificial Intelligence systems during research execution:

- **Search & Literature Discovery**: `DeepSeek` and `Google Gemini` were used for technical reference searching and exploratory querying; `STORM`, `Consensus`, and `ResearchRabbit` were employed for citation graphing and mapping connected papers.
- **Writing & LaTeX Formatting Assistance**: `DeepSeek` and `Google Gemini` were utilized to assist with section drafting, LaTeX code styling, and manuscript redaction. All conceptual arguments, analytical protocols, data checks, and manuscript text were independently verified, reviewed, and finalized by the authors.

Data and Code Availability

Complete pipeline scripts, split indices, and reproducibility notebooks will be published upon publication release.

Acknowledgments. Supported by SUP'RH School of Management & AI, Morocco.

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